

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

595040065 - Women In Science And Technology

DEGREE PROGRAMME

59ET - Doble Grado En Ing.Electronica De Comunicaciones Y En Ing.Telematica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	595040065 - Women In Science And Technology
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	Third year
Semester of tuition	Semester 6
Tuition period	February-June
Tuition languages	English
Degree programme	59ET - Doble Grado en Ing.electronica de Comunicaciones y en Ing.telematica
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Maria Pilar Ochoa Perez (Subject coordinator)	A3111	pilar.ochoa@upm.es	Sin horario. Tutoring hours will be published on Moodle.
Amador Miguel Gonzalez Crespo	A3111	amador.m.gonzalez@upm.es	Sin horario. Tutoring hours will be published on Moodle

Marta Sanchez Agudo	A3111	marta.sanchez@upm.es	Sin horario. Tutoring hours will be published on Moodle.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

2.3. External faculty

Name and surname	Email	Institution
Margarita Martínez Núñez	margarita.martinez@upm.es	ETSIST - Ing. de Organización, Administración de Empresas y Estadística
Ana Belén García Hernando	anabelen.garcia@upm.es	ETSIST - Ingeniería Telemática y Electrónica
María Luisa Martín Ruiz	marialuisa.martinr@upm.es	ETSIST - Ingeniería Telemática y Electrónica

3. Skills and learning outcomes *

3.1. Skills to be learned

CG 02 - Capacidad de búsqueda y selección de información, de razonamiento crítico y de elaboración y defensa de argumentos dentro del área.

CG 03 - Capacidad para expresarse correctamente de forma oral y escrita y transmitir información mediante documentos y exposiciones en público.

CG 05 - Capacidad de trabajo en equipo y en entornos multidisciplinares.

CG 11 - Habilidades para la utilización de las Tecnologías de la Información y las Comunicaciones.

CG 12 - Habilidad para las relaciones interpersonales y el trabajo en un contexto nacional e internacional, con capacidad para expresarse de forma oral y escrita en lengua inglesa.

CG 14 - Actitudes de ética y responsabilidad profesional, respeto a los Derechos Humanos y a la diversidad cultural.

3.2. Learning outcomes

RA286 - 194 - Podrá procesar la información de diferentes fuentes y sintetizarla adecuadamente

RA956 - Identificar y valorar cuestiones éticas relativas al desarrollo tecnológico

RA957 - Utilizar software de apoyo a las presentaciones

RA958 - Cumplir con los plazos preestablecidos

RA959 - Organizar información obtenida de textos breves

RA960 - Comunicar resultados oralmente

RA961 - Exponer ideas de forma oral o escrita en lengua inglesa

RA962 - Trabajar en proyectos compartidos

RA955 - - Valorar y discutir las consecuencias sociales de desarrollos tecnológicos

RA283 - 195 - Podrá contrastar las ideas aportadas por diferentes fuentes

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This course provides a basic overview of the history of women in science and technology. The specific contributions of women in a variety of disciplines will be studied while attention will also be given to the various scientific and technical concepts necessary to understand these contributions. In this way, we also aim to increase the scientific and technical background of the students. The class will also address how both historical and modern biases within science and technology, as well as in the portrayals of women and girls in the media and popular culture, can affect outcomes in these areas.

The development of the course syllabus described below will not be done in a rigid way. The different topics will be

covered in the form of talks, debate and various interactive activities that may cover different aspects of the course at the same time.

Students from all fields and levels of preparation are welcome.

4.2. Syllabus

1. Introduction.

- 1.1. What do we understand by science and gender?
- 1.2. Motivation of this subject in the current sociocultural context.

2. History of women in science and technology

- 2.1. Women scientists in the ancient world and Middle Ages
- 2.2. From the Enlightenment to the 19th century
- 2.3. The 19th and early 20th centuries
- 2.4. World War II and social changes

3. Revolutionary scientists, but not recognized.

- 3.1. The Matilda effect and its consequences.
- 3.2. The rarity of female Nobel laureates.

4. Current role of women in science.

5. How sexism and stereotyping in vocational education promote and reinforce gendered occupations

6. Neurosexism: the myth that men and women have different brains

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Course introduction, instructor introduction, student introductions. Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15 Active participation of students during lectures, contributing to debates will be taken into account in their assessment. Other assessment Progressive assessment Presential Duration: 00:00
2	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
3	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			
4	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15

	Participation in class Duration: 00:00 Additional activities			
5	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
6	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
7	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
8	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
9	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15

	Additional activities			
10	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
11	Presentation of a topic and in-class discussion. Duration: 01:45 Additional activities Daily short quiz. Duration: 00:15 Additional activities Participation in class Duration: 00:00 Additional activities			Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed Other assessment Progressive assessment Presential Duration: 00:15
12	Presentation of a topic and in-class discussion: students class presentations. Duration: 02:00 Additional activities Participation in class Duration: 00:00 Additional activities			Oral presentation in class by all members of the group. Each group of students will present their work on a day in either week 12 or week 13. Group presentation Progressive assessment Presential Duration: 00:15
13	Presentation of a topic and in-class discussion: students class presentations. Duration: 02:00 Additional activities Participation in class Duration: 00:00 Additional activities			Oral presentation in class by all members of the group. Each group of students will present their work on a day in either week 12 or week 13. Group presentation Progressive assessment Presential Duration: 00:15 Students will work in groups on a topic and prepare a poster, leaflet or similar related to the topic. Group work Progressive assessment Not Presential Duration: 00:00
14				
15				
16				
17				Global Examination: Written Test (12-06-2026) Written test Global examination Presential Duration: 00:45 Global Examination: Oral presentation

				(12-06-2026) Individual presentation Global examination Presential Duration: 00:20
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
1	Active participation of students during lectures, contributing to debates will be taken into account in their assessment.	Other assessment	Face-to-face	00:00	15%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
2	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
4	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
5	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 05 CG 11 CG 12 CG 14 CG 02 CG 03

6	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
7	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
8	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
9	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
10	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
11	Daily short quiz on the topic of the day's lesson/equivalent assessable activities. These activities are daily carried out in class and their duration will depend on the specific task to be performed	Other assessment	Face-to-face	00:15	20%	/ 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
12	Oral presentation in class by all members of the group. Each group of students will present their work on a day in either week 12 or week 13.	Group presentation	Face-to-face	00:15	35%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14

13	Oral presentation in class by all members of the group. Each group of students will present their work on a day in either week 12 or week 13.	Group presentation	Face-to-face	00:15	35%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
13	Students will work in groups on a topic and prepare a poster, leaflet or similar related to the topic.	Group work	No Presential	00:00	30%	3 / 10	CG 05 CG 11 CG 12 CG 14 CG 02 CG 03

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Global Examination: Written Test (12-06-2026)	Written test	Face-to-face	00:45	35%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
17	Global Examination: Oral presentation (12-06-2026)	Individual presentation	Face-to-face	00:20	30%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Referred examination: Written Test (30-06-2026)	Written test	Face-to-face	01:00	35%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14
Referred examination: Oral presentation (30-06-2026)	Individual presentation	Face-to-face	00:30	30%	3 / 10	CG 02 CG 03 CG 05 CG 11 CG 12 CG 14

6.2. Assessment criteria

The specific dates of Global Examination and Referred Examination depend on the organization of the School's exam schedule, coordinated by the SOA, and are published in the Annual Teaching Plan (Plan Anual Docente). In case of discrepancy between the information published in this guide and that published in the Annual Teaching Plan, the information published in the Annual Teaching Plan should be taken into account, as it contains the appropriate updates.

Progressive Assessment

* Daily short quiz/equivalent assessable activities on the topic of the day's lesson. Several tests/activities will be held daily in the course. The average mark obtained in these activities accounts for 20 % of the total grade (a test not taken counts as zero for averaging purposes). It is not possible to carry out a second-chance examination for these activities because their aim is to assess the student's contribution during the face-to-face lessons and group work.

* Active participation of students during lectures. Contributing to debates will be taken into account in the assessment. Student participation in debates and discussions accounts for 15% of the total grade. It is not possible to carry out a second chance examination for these activities because their aim is to assess the student's participation in face to face debates.

* Students will work in groups on a topic (proposed by the teachers or by the students themselves and approved by the teachers). They will have to present information on the chosen topic in two formats:

(a) poster, brochure or similar (this task accounts for 30% of the total grade).

(b) oral presentation in class by all members of the group (this presentation accounts for 35% of the total grade).

To pass the course, the student is required to meet all the following conditions:

* Obtain a minimum grade of 3 points (out of 10) in each of the two tasks marked (a) and (b) above.

* Obtain a minimum grade of 5 points (out of 10) in the global mark of the course.

Students who do not meet any of the previous conditions will not be able to obtain a grade higher than 4.0 points (out of 10) in the course.

Global examination

Only those students who have not passed the progressive assessment will have to take the Global Examination. Students will be evaluated based on:

- * Daily short quiz/equivalent assessable activities on the topic of the day's lesson. Several tests/activities will be held daily in the course. The average mark obtained in these activities accounts for 20 % of the total grade (a test not taken counts as zero for averaging purposes). It is not possible to carry out a second-chance examination for these activities because their aim is to assess the student's contribution during the face-to-face lessons and group work.

- * Active participation of students during lectures. Contributing to debates will be taken into account in the assessment. Student participation in debates and discussions accounts for 15% of the total grade. It is not possible to carry out a second chance examination for these activities because their aim is to assess the student's participation in face to face debates.

- * Oral presentation (accounts for 30% of the total mark)

- * Written test (accounts for 35% of the total mark)

To pass the course, the student is required to meet all the following conditions:

- * Obtain a minimum grade of 3 points (out of 10) in each of the two tasks, oral presentation and written test.

- * Obtain a minimum grade of 5 points (out of 10) in the global mark of the course.

Students who do not meet any of the previous conditions will not be able to obtain a grade higher than 4.0 points (out of 10) in the course.

Referred examination (July)

- * Daily short quiz/equivalent assessable activities on the topic of the day's lesson. Several tests/activities will be held daily in the course. The average mark obtained in these activities accounts for 20 % of the total grade (a test not taken counts as zero for averaging purposes). It is not possible to carry out a make-up exam for these activities because the aim is to assess the student's contribution to the development of the class and face to face group work.

* Active participation of students during lectures, contributing to debates will be taken into account in their assessment. Student participation in debates and discussions account for 15% of the total grade. It is not possible to carry out a make-up exam for these activities because the aim is to assess the student's participation in debates.

* Oral presentation (accounts for 30% of the total mark)

* Written Test (accounts for 35% of the total mark).

To pass the course, the student is required to meet all the following conditions:

* Obtain a minimum grade of 3 points (out of 10) in each of the two tasks, oral presentation and written test.

* Obtain a minimum grade of 5 points (out of 10) in the global mark of the course.

Students who do not meet any of the previous conditions will not be able to obtain a grade higher than 4.0 points (out of 10) in the course.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Has feminism changed science?. Londa Schiebinger. Harvard University Press, 1999	Bibliography	
Women in Science: a Social and Cultural History. Ruth Watts. Routledge, 2007	Bibliography	
Notable Women in the Physical Sciences. A Biographical Dictionary. Barbara Smith Shearer and Benjamin F. Shearer. ABC-CLIO, 1997.	Bibliography	
Women in science. European Commision. Publications Office of the European Union, 2010	Bibliography	
Madame Curie: A Biography. Eve Curie. Da Capo Press, 2001.	Bibliography	
The biographical dictionary of women in science: pioneering lives from ancient times to the mid-20th century. Marilyn Ogilvie and Joy Harvey. Routledge, 2000.	Bibliography	
Inferior: How science got women wrong -and the new research that?s rewriting the story. A. Saini. Beacon Press.	Bibliography	
Delusions of Gender. Cordelia Fine. London icon books, 2011.	Bibliography	

The Gendered Brain: The new neuroscience that shatters the myth of the female brain. Gina Rippon. Bodley Head, 2019.	Bibliography	
Moodle	Web resource	
Classroom equipment	Equipment	Personal computer. Video projector. Blackboard and Digital Blackboard
Recoding Gender- Women's Changing Participation in Computing. Janet Abbate. Massachusetts Institute of Technology, 2012.	Bibliography	

8. Other information

8.1. Other information about the subject

>Sustainable Development Goals

The course aims to contribute, mainly, to the development of the following SDGs:

* SDG 4: Quality Education

It goes without saying that our teaching activity, like that of the entire University, is focused on guaranteeing a Quality Education to our students. To this end, we pay close attention to the content of each class as well as its distribution and development, we prepare the necessary material in each case and we provide the students with all the information and support necessary to ensure the quality of their education. We consider it interesting to point out the following SDG 4 targets (<https://www.un.org/sustainabledevelopment/education/>), which are particularly relevant to this specific course:

4.3 By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university

4.5 By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous people and children in vulnerable situations

4.7 By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development

*** SDG 5: Gender Equality**

Given the content of this course, it is beyond doubt that there is a strong link to the SDG 5. More specifically, we can point out the following targets (<https://www.un.org/sustainabledevelopment/gender-equality/>):

5.1 End all forms of discrimination against all women and girls everywhere

5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decisionmaking in political, economic and public life

5.B Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women

>Learning Skills:

CG 02 - Ability to search and select information, critical reasoning and elaboration and defense of arguments within the area.

CG 03 - Ability to express oneself correctly orally and in writing form to communicate information through documents and public presentations.

CG 05 - Ability to work in a team and in multidisciplinary environments.

CG 11 - Skills for the use of the Information and Communication Technologies.

CG 12 - Ability for interpersonal relations and work in a national and international context, with the capacity to express oneself orally and in writing form in the English language.

>Communication with the teachers:

The student must request the tutorship by appointment from the teacher of the group in which he or she is enrolled. This can be done in person (when possible) or by email. The tutoring schedule for each teacher will be posted in Moodle at the beginning of the semester.

The information contained in this guide is indicative, and is therefore subject to change due to errors, omissions, unforeseen incidents or if the correct development of the course makes it advisable to do so.

The timetable given in this guide is orientative, as it is designed taking into account thirteen nominal teaching weeks. Therefore, students should consult the corresponding timetable in the School's Annual Teaching Plan.